

Boundary-spanning search and breakthrough innovation: the moderating role of big data analytics capability

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Abstract

Purpose – The rapid advancement of digital transformation requires a shift in firms' focus from past met needs to both latent future and unmet past needs. However, how boundary-spanning search with future orientation and past orientation affects breakthrough innovation remains unclear. This study thus aims to investigate the relationship between boundary-spanning search and breakthrough innovation from the perspective of search orientation.

Design/methodology/approach – In terms of search orientation, this study divides boundary-spanning search into forward-looking search and backward-looking search. Drawing on resource-based view, this study develops a theoretical model in which big data analytics capability moderates the effects of forward-looking and backward-looking searches on breakthrough innovation. Empirical analyses were conducted on data from China's advanced manufacturing firms. Research model and hypotheses were tested through multiple regression.

Findings – The results confirm that forward-looking search has a positive effect on breakthrough innovation, and big data analytics capability strengthens this positive effect. Furthermore, backward-looking search has an inverted U-shaped effect on breakthrough innovation. Interestingly, as big data analytics capability increases, this inverted U-shaped curve flattens and becomes almost linear.

Originality/value – This study uncovers the different effects of boundary-spanning search with different orientations on breakthrough innovation and extends the research on the relationship between boundary-spanning search and breakthrough innovation by incorporating search orientation. Furthermore, by demonstrating the moderating role of big data analytics capability, this study provides a crucial condition under which boundary-spanning search can enhance breakthrough innovation.

Keywords Boundary-spanning search, Big data analytics capability, Breakthrough innovation, Digital transformation

Paper type Research article

1. Introduction

With the rapid advancement of digital transformation, there emerge a variety of new business scenarios, such as virtual industrial parks, smart factories and digital trade, giving rise to new market needs. Furthermore, digital technologies have the characteristics of frequent iteration and update, which position firms in a fast-changing environment. Although focusing on unmet past needs can effectively accelerate firms' breakthrough innovation, it is limited to the maintaining of current competitive advantages (Wang *et al.*,



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2023). To gain success in the fast-changing environment, it is also vital to develop corporate foresight and proactively predict technology trends and market needs (Marinković *et al.*, 2022). This requires a shift in firms' innovation focus from unmet past needs to both latent future and unmet past needs.

Boundary-spanning search refers to firms' knowledge search activities that cross organizational or technical boundaries (Lee *et al.*, 2024). It can access to heterogenous and novel knowledge, perspectives and ideas to think outside the box and generate creative business insights (Acharya *et al.*, 2022; Menguc *et al.*, 2014). Therefore, it has been a critical driver of breakthrough innovation. In terms of orientations to latent future needs and unmet past needs, boundary-spanning search can be divided into forward-looking search and backward-looking search. Specifically, forward-looking search refers to the process by which firms seek knowledge beyond their borders to anticipate and satisfy latent future needs (Jissink *et al.*, 2019). Firms that conduct forward-looking search frequently scan for external knowledge to capture market trends and potential opportunities for new products. Backward-looking search refers to the process by which firms seek knowledge beyond their borders to meet needs not met in the past (Chen, 2008; Zan *et al.*, 2022). Firms that conduct backward-looking search constantly look for external knowledge to improve existing products and processes.

Existing studies have explored the effect of boundary-spanning search on breakthrough innovation from various perspectives. From the perspective of knowledge type, Jung and Lee (2016) found the positive relationship between original knowledge search and breakthrough innovation. From the perspective of search scope, Wang *et al.* (2020) identified the inverted U-shaped effects of search depth and search breadth on firms' innovation. Furthermore, from the perspective of search timing, Yang *et al.* (2021) demonstrated the mechanism of how proactive search and responsive search influence firms' innovation. However, few studies explored the relationship between boundary-spanning search and breakthrough innovation from the perspective of search orientation. Since a firm's strategic orientation largely determines how it conduct and manage its business activities (Spanjol *et al.*, 2012), the search orientations to latent future needs and unmet past needs may result in different search efforts and outcomes. Therefore, it is necessary to investigate how forward-looking search and backward-looking search affect breakthrough innovation.

Moreover, recent studies highlighted big data analytics capability as an accelerator of firms' innovation processes (Al-Omouh *et al.*, 2024; Bag *et al.*, 2024; Kumar *et al.*, 2021). Big data analytics capability is the ability to quickly process vast quantities of data to provide firms with timely environment information (Al-Omouh *et al.*, 2024). The continuously advancement of digital transformation generates unprecedented volume, variety and velocity data, namely big data, increasing the demand for data analytics (Nisar *et al.*, 2021). Firms with higher big data analytics capabilities can extract valuable information from large amounts of data sets to capture consumer needs and market trends (Hyun *et al.*, 2023). Such information is vital for firms to leverage the knowledge obtained through boundary-spanning search to create novel insights, identify emerging opportunities, and make informed decisions (Lozada *et al.*, 2023). That is, big data analytics capability may moderate the effects of boundary-spanning on breakthrough innovation. However, whether and how it accelerates the transformation from boundary-spanning search to breakthrough innovation is still unclear.

To fill the above research gaps, this study investigates how boundary-spanning search affects breakthrough innovation from the perspective of search orientation and examine the moderating role of big data analytics capability. In terms of firms' search orientations to latent future needs and unmet past needs, this study divides boundary-spanning search into forward-looking search and backward-looking search (Chen, 2008; Jissink *et al.*, 2019). Drawing upon resource-based view (RBV), this study develops a theoretical model in which

big data analytics capability moderates the effects of forward-looking search and backward-looking search on breakthrough innovation. To examine the theoretical model, this study conducts empirical analyses on data from China's advanced manufacturing firms. Results from multiple regression confirm the positive effect of forward-looking search on breakthrough innovation, and big data analytics capability strengthen this positive effect. Additionally, backward-looking search has an inverted U-shaped effect of breakthrough innovation, and big data analytics capability flattens this inverted U-shaped curve.

The main contributions of this study are as follows. First, from the perspective of search orientation, this study divides boundary-spanning search into forward-looking search and backward-looking search, and examines their effects on breakthrough innovation. Although prior research examined the effects of boundary-spanning search on breakthrough innovation from various perspectives (Wang *et al.*, 2020; Yang *et al.*, 2021), little has considered the search orientation deriving from the advancement of digital transformation. By considering search orientation, this study enriches the literature of boundary-spanning search, advances the research on the relationship between boundary-spanning search and breakthrough innovation, and extends RBV. Second, this study identifies the different influences of forward-looking and backward-looking searches on breakthrough innovation. In this way, this study uncovers the different effects of boundary-spanning search with different orientations on breakthrough innovation. Third, by demonstrating the moderating role of big data analytics capability, this study sheds light on the underlying mechanism by which forward-looking and backward-looking searches affect breakthrough innovation. Hence, this study echoes the arguments that big data analytics capability play a critical role in firms' innovation (Bag *et al.*, 2024; Kumar *et al.*, 2022). It also provides a crucial condition under which boundary-spanning search can promote breakthrough innovation in the digital era.

The remainder of this paper is organized as follows. Section 2 presents theoretical background and hypotheses. Section 3 is the research design, including sample and data collection, variables and measures, bias and common method variance, reliability and validity. Section 4 reports the results of empirical analyses. Section 5 discusses the results, theoretical and managerial implications. Section 6 presents the conclusions, research limitation and future research.

2. Theoretical background and hypotheses

The literature about boundary-spanning search, big data analytics capability, and breakthrough innovation was acquired from WOS and CNKI databases. By analyzing relevant literature through Citespace software (Chen, 2006), we induce the definitions, dimensions, antecedents and research trends of each variable, and identify the foundation theory, namely RBV. According to RBV, resources, especially knowledge resources, are the critical sources of firms' sustainable competitive advantage. Boundary-spanning search is an important way to access to various knowledge which in turn can foster firms' innovation, and subsequently promote sustainable competitive advantage (Lee *et al.*, 2024; Yang *et al.*, 2021). Therefore, based upon RBV, we develop a theoretical model to investigate the mechanism through which boundary-spanning search affects breakthrough innovation.

2.1 Boundary-spanning search and breakthrough innovation

2.1.1 Forward-looking search and breakthrough innovation. Forward-looking search refers to the process by which firms seek knowledge beyond their borders to anticipate and satisfy latent future needs. Firms that conduct forward-looking search frequently scan for external knowledge to capture market trends and potential opportunities for new products. The

influence of forward-looking search on breakthrough innovation can be manifested in three ways. First, forward-looking search promotes proactive strategy planning for breakthrough innovation. According to organizational learning theory, forward-looking search may acquire external knowledge which involves environment changes and forthcoming trends. Such knowledge is the basis to formulate proactive strategy for new product development, new market entry and new technology adoption (Bhatia, 2021). As business strategy theories suggest, proactive strategy is a medium that figures out firms' long-term direction and shared goals. Under the guidance of proactive strategy, firms are able to develop plans about future scenarios (Marinković *et al.*, 2022). While scenario planning determines how to integrate various resources, shared goals promote coordinate multiple innovation activities (Patricio *et al.*, 2020). In this way, firms could effectively deploy and utilize various innovation resources, thus enhancing breakthrough innovation.

Second, forward-looking search generates novel insights to initiate breakthrough innovation. Firms that conduct forward-looking search usually scan for the knowledge about customer requirements, emerging technologies and competitors' actions (Jissink *et al.*, 2019). While constantly scanning for customer requirements allows firms to capture the trends in customer needs and anticipate future customer needs, focusing on emerging technologies enables firms to utilize novel and powerful technologies to meet new customer needs (Marinković *et al.*, 2022; Schwarz *et al.*, 2020). Additionally, by monitoring competitors' R&D projects, patenting programs, and new product launch activities, forward-looking search may acquire heterogeneous knowledge and create different ideas about new products and service ways (Yang *et al.*, 2021).

Third, forward-looking search challenges conventional routines to enhance breakthrough innovation. By aiming at future needs, forward-looking search may bring in valuable knowledge to deepen firms' understanding about future and help to develop corporate foresight (Schwarz *et al.*, 2020). When the perceived signals of breakthrough innovation do not fit firms' expectations, a deep understanding about future may restructure firms' interpretation routines (Jissink *et al.*, 2019; Rohrbeck and Gemünden, 2011). This encourages firms to interpret innovation signals in alternatives ways, and enables firms to seize innovation opportunities. On the other hand, as Marinković *et al.* (2022) suggested, firms who have corporate foresight are more capable to create agile R&D routines. Such agile routines are critical for firms to overcome managerial myopias, engage in under-explored domains and capture weak signals of breakthrough innovation.

Based upon the above arguments, this study hypothesizes:

H1a. Forward-looking search has a positive effect on breakthrough innovation.

2.1.2 Backward-looking search and breakthrough innovation. Backward-looking search refers to the process by which firms seek knowledge beyond their borders to meet needs not met in the past. Firms that conduct backward-looking search constantly look for external knowledge to improve existing products and processes (Chen, 2008; Zan *et al.*, 2022). According to behavioral theory, the discrepancy between companies' actual performance and aspirations affects companies' decision-making in R&D activities. Agency theory further suggests that, companies generally take actions to minimize the potential negative outcomes that could harm their performance. To a certain extent, unmet needs reflect the discrepancy between the actual effectiveness and aspirations of firms' products and processes, and may lead to negative performance. Given that breakthrough innovation can effectively improve firms' products and processes, unmet needs may be important triggers for breakthrough innovation (Lv *et al.*, 2019). In this logic, backward-looking search may strengthen firms' motivation to obtain nonlocal knowledge for new product development and business process revolution. Additionally, unmet needs provide a negative performance frame for decision makers. Based on financial theory, firms' attitudes toward risk will shift from risk averse to

risk seeking (Fehle and Tsyplakov, 2005). That is, firms may accept risks to avoid negative performance. Hence, firms that conduct backward-looking search may have higher risk tolerance and deploy the obtained knowledge into more radical R&D activities (Hu *et al.*, 2022).

Besides the above positive effects, backward-looking search also has some limitations. First, by focusing on unmet past needs, backward-looking search may obtain obsolete knowledge which cannot support breakthrough innovation. The advancement of digitalization continuously creates new digital technologies and application scenarios, positioning firms in a rapidly changing environment. To adapt to technological changes, firms should quickly capture market trends and predict future needs to conduct breakthrough innovation, so as to enhance their agility and dynamic capability (Al-Omoush *et al.*, 2024). However, unmet past needs may have been addressed by outdated technologies or methodologies that are no longer relevant or effective in the current market. Hence, the knowledge obtained through backward-looking search may be outdated to respond to the ever-changing environment and achieve valuable breakthrough innovation. Second, backward-looking search pays more attention to the knowledge about specific problems of products and processes, restricting firms' search scope (Posen *et al.*, 2018). Focusing on specific problems may consume significant resources in terms of time, human resources, and finances. According to attention theory, there may be fewer resources available for exploring alternative knowledge or new opportunities outside the scope of these problems (Brielmaier and Friesl, 2023). Given that breakthrough innovation relies more heavily on various and heterogeneous knowledge, the limited knowledge may decrease the likelihood of generating path-breaking novelty (Jung and Lee, 2016). Third, as Angus (2019) argued, searched knowledge about a given problem is less applicable in a dissimilar context. Hence, when exploring a new domain or developing new products, backward-looking search may be unable to generate adequate innovation outcomes.

A possible explanation for these inconsistent arguments may be that extant research has not taken the level of backward-looking search into consideration. Specifically, as the levels of backward-looking search increase from low to moderate, firms' risk tolerance and motivation to breakthrough innovation increase (Hu *et al.*, 2022). This will encourage firms to venture the obtained knowledge into more radical R&D activities (Lv *et al.*, 2019). Although there may come obsolete and limited knowledge reducing the novelty of breakthrough innovation, firms can overcome these disadvantages through continuously environment-scanning and knowledge integration activities (Yang *et al.*, 2021). Thus, from low to moderate levels, the positive effect of backward-looking search on breakthrough innovation is greater than its negative effect. However, as the levels of backward-looking search increase from moderate to high, firms have to spend more efforts and costs to reduce the negative impact deriving from the decreasing innovation novelty (Jung and Lee, 2016). As a result, the negative effect of backward-looking search will gradually outweigh its positive effect and eventually be dominant in the relationship between backward-looking search and breakthrough innovation. Therefore, this study hypothesizes:

H1b. Backward-looking search has an inverted U-shaped effect on breakthrough innovation.

2.2 The moderating role of big data analytics capability

Big data analytics capability refers to the ability to quickly process vast quantities of data to provide firms with timely environment information. The influence of big data analytics capability on the relationship between forward-looking search and breakthrough innovation can be manifested as follows. First, big data analytics capability enables informed strategy plans. Due to the character of evidence- and data-based, big data analytics is with less human

bias (Mikalef *et al.*, 2019). Hence, firms with higher big data analytics capabilities can conduct complex analysis on various types of data, and extract accurate information about future (Ansari and Ghasemaghaei, 2023). Such information enables firms to make use of forward-looking search to develop more informed strategy plans for breakthrough innovation (Lozada *et al.*, 2023). By enabling informed strategy plans, big data analytics capability may improve the likelihood of transforming forward-looking search into breakthrough innovation.

Second, big data analytics capability increases creative idea identification. With assistance of advanced algorithms, big data analytics can collect and analyze vast amounts of data from various sources (Mikalef *et al.*, 2019). Through these analyses, firms may uncover novel business patterns that go beyond simple and traditional correlations (Lozada *et al.*, 2023). Furthermore, big data analytics can also recognize anomalies in the data, which might indicate potential opportunities for innovation (Habeeb *et al.*, 2019). These patterns and opportunities provide more creative ideas for firms to leverage the knowledge obtained by forward-looking search to achieve breakthrough innovation. In other words, by identifying creative ideas, big data analytics capability may speed the transformation from forward-looking search to breakthrough innovation.

Third, big data analytics capability facilitates cross-functional collaboration. Big data analytics can analyze data from different departments and teams, identify the common interest among functions, and indicate collaboration opportunities. This facilitates insights and knowledge sharing among marketing, R&D, and design teams, and encourages synthesis of diverse ideas and perspectives. In this way, firms with higher big data analytics capability are able to fully interpret innovation signals, and efficiently deploy the knowledge obtained by forward-looking search to promote breakthrough innovation (Ansari and Ghasemaghaei, 2023; Dubey *et al.*, 2019). Therefore, by facilitating cross-functional collaboration, big data analytics capability may increase the efficiency in transforming forward-looking search into breakthrough innovation.

Based upon the above arguments, this study hypothesizes:

H2a. Big data analytics capability moderates and strengthens the positive effect of forward-looking search on breakthrough innovation.

The influence of big data analytics capability on the relationship between backward-looking search is twofold. On the one hand, as backward-looking search increase from low to moderate level, big data analytics capability strengthens its positive effect on breakthrough innovation. As Lozada *et al.* (2023) suggested, analyzing data on external factors such as market volatility, technology trends and customer behavior can help firms to predict potential risks. Accordingly, firms with higher big data analytics capability can proactively develop risk management strategies to increase risk tolerance and carry out more radical innovation projects. The other advantage of big data analytics is that it can use various types of data to identify emerging innovation opportunities (Kumar *et al.*, 2021). When perceiving opportunities, firms will be encouraged to set ambitious goals to achieve innovation (Chen and Liu, 2020). In other words, by identifying innovation opportunities, big data analytics capability increases firms' motivation to breakthrough innovation. Taken together, big data analytics capability can enhance firms' motivation and risk tolerance in leveraging backward-looking search to achieve breakthrough innovation.

On the other hand, as backward-looking search increase from moderate to high level, big data analytics capability mitigates its negative effect on breakthrough innovation. When the level of backward-looking search is high, firms may devote their most efforts to search for knowledge about unfulfilled needs. As mentioned before, high-level of backward-looking search may lead to obsolete and limited knowledge, decreasing innovation novelty. As Habeeb *et al.* (2019) argued, big data analytics can provide a rapid feedback loop by monitoring real-time data. This allows firms to detect environment changes, identify new

trends, and test their innovative ideas quickly (Kumar *et al.*, 2021). Hence, higher big data analytics capability enables firms to reduce the obsolescence risks and acquire updated knowledge. Moreover, big data analytics is positively related to firms' partnering agility (Hyun *et al.*, 2023). Firms with higher big data analytics capability can flexibly leverage the knowledge from their partners, including customers, suppliers, distributors and so on (Kumar *et al.*, 2021). In this sense, big data analytics capability extends firms' search scope, and allows them to access to multiple knowledge about unfulfilled needs, thus increasing innovation novelty.

Therefore, this study hypothesizes:

H2b. Big data analytics capability moderates and flattens the inverted U-shaped effect of backward-looking search on breakthrough innovation.

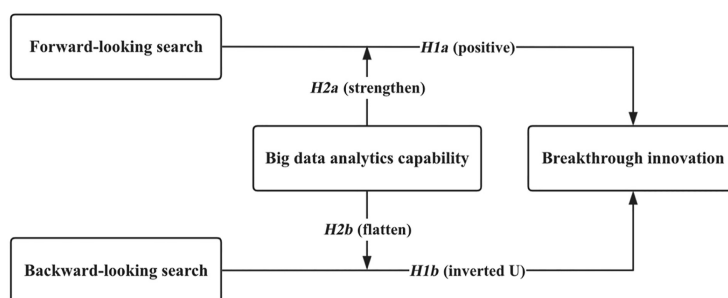
Figure 1 shows the theoretical model.

3. Research design

3.1 Sample and data collection

This study selected China's advanced manufacturing firms as the research sample, which include intelligent manufacturing equipment, marine engineering equipment, aerospace equipment, new materials, new energy vehicles and so on. First, China has a huge manufacturing system, among which advanced manufacturing industries are the most active field of innovation (Yang *et al.*, 2021). Firms in these industries invest more efforts to improve innovation capabilities. Second, digital technologies are widely adopted in advanced manufacturing industries (Yang *et al.*, 2021; Zhao *et al.*, 2022). Sampling firms in these industries can deeply understand the influence of digital transformation. Third, the government of the People's Republic of China published the 2035 Long-term Goal Outline. It emphasizes the important strategic role of advanced manufacturing in enhancing China's industrial capabilities, and provides many support policies.

This study collected data from advanced manufacturing firms in Shenzhen, Suzhou, Xi'an, Zhengzhou, Weifang and Hengyang. According to the Report on China's Top 100 Advanced Manufacturing Cities in 2022, the development of advanced manufacturing firms can be divided into four stages. These stages are the leapfrogging, expansion, developing, and start-up stages. Firms in the leapfrogging stage have established solid manufacturing bases and industrial competitiveness, and are mainly located in Shenzhen, Suzhou and other East China regions. Firms in the expansion stage have developed multiple industrial chains, but there is still a gap with the first level. They are mainly located in Xi'an and Zhengzhou. Firms in the developing stage have gained strong competitive advantages in the subdivision of industrial



Source(s): Authors work

Figure 1.
Theoretical model

chains, and have great potential for future development. They are mainly located in the sub-central cities in East and Central China, such as Weifang. Firms in the start-up stage have good traditional manufacturing bases, but the development of advanced manufacturing field is in its infancy. They are mainly located in Hengyang and other cities. Therefore, firms in these six cities can represent the different levels of advanced manufacturing firms in China.

To guarantee the typicality of samples, we randomly selected 150 firms in each city according to the local business directory, a total of 900 firms. Moreover, to conduct empirical research, we adapted Gerbing and Anderson's (1988) three-step procedure. This procedure is a methodological approach to address issues related to research sample and data collection, including measurement scales, measurement modification and data collection. Specifically, we firstly developed an English questionnaire based upon extant scales. To guarantee the conceptual equivalence, we performed the double- and back-translation. Second, to ensure the validity of questionnaire, we interviewed five managers and three scholars related to advanced manufacturing industries. Moreover, we also performed a pretest with 58 MBA students in advanced manufacturing firms. According to manager and scholars' suggestions and protest results, we modified and reworded several items, and formed the final questionnaire. Third, the data collection period was from February to May 2023. We invited the selected firms to participate in the survey through telephone and email, and 416 firms agreed. Given that the directors of R&D department have a broad view on firms' innovation activities (Atuahene-Gima K, 2005), we issued the final questionnaire to them. After several reminders, we received 352 responses and finally got 314 usable responses by excluding invalid ones, generating an effective response rate of 34.9%. Table 1 presents the characteristics of samples.

Profiles	Frequency	Sample (N = 314)	Percentage
<i>Firm size (employees)</i>			
<500	45		14.3
501–1,000	79		25.2
1,001–1,500	98		31.2
1,501–2000	66		21.0
>2000	26		8.3
<i>Firm age (years)</i>			
<5	29		9.2
5–10	68		21.7
11–15	100		31.8
16–20	83		26.4
>20	34		10.8
<i>R&D intensity (%)</i>			
<5	25		8.0
5–10	66		21.0
11–15	119		37.9
16–20	84		26.8
>20	20		6.4
<i>Digitalization experience (years)</i>			
<2	29		9.2
2–4	66		21.0
5–7	108		34.4
8–10	75		23.9
>10	36		11.5

Table 1.
Sample characteristics Source(s): Authors' work

3.2 Variables and measurement

The measures used in this study are listed in Table 2, and each item were operationalized on a 5-point Likert scale.

3.2.1 Boundary-spanning search. In terms of search orientation, this study divides boundary-spanning search into forward-looking search and backward-looking search. Drawing on the research of Jissink *et al.* (2019) and Chen (2008), this study used a four-item scale to measure forward-looking search. For backward-looking search, it was measured by the four-item scale adapted from Chen (2008) and Zan *et al.* (2022).

3.2.2 Big data analytics capability. Big data analytics capability refers to the ability to quickly process vast quantities of data to provide firms with timely environment information. In line with the research of Yu *et al.* (2021), this study used a four-item scale to measure it.

3.2.3 Breakthrough innovation. Breakthrough innovation is a revolutionary activity that emphasizes the creation of a new business line. According to the research of Menguc *et al.* (2014), a four-item scale were used to reflect breakthrough innovation.

3.2.4 Control variables. First, we controlled for firm size as it influences a firm's innovation efficiency, and measured it by using the number of employees (Mikalef *et al.*, 2019). Second, since firm age has influences on innovation strategies, we considered it as a control variable and measured it by using the years a firm has existed (Yang *et al.*, 2021). Third, Menguc *et al.* (2014) found that R&D intensity has positive effects on breakthrough innovation. Therefore,

Items	Loadings
<i>Forward-looking search (FLS)</i> ($\alpha = 0.822$, AVE = 0.653, CR = 0.824)	
FLS1. Our company continuously scans for external knowledge to capture key trends and potential opportunities for new products	0.808
FLS2. Our company frequently tracks competitors' trends, policies, and tactics relevant to our new products	0.843
FLS3. Our company constantly looks for short-term and long-term changes which could impact new products	0.783
FLS4. Our company regularly seeks for future threats that could come up for new products	0.796
<i>Backward-looking search (BLS)</i> ($\alpha = 0.839$, AVE = 0.674, CR = 0.839)	
BLS1. Our company often looks for various knowledge to solve existing management problems	0.809
BLS 2. Our company is committed to leveraging external knowledge to improve existing technologies	0.856
BLS 3. Our company constantly leverages external knowledge to fine-tune existing production processes	0.789
BLS 4. Our company frequently searches various knowledge to enrich existing knowledge reserves	0.829
<i>Big data analytics capability (BDAC)</i> ($\alpha = 0.935$, AVE = 0.837, CR = 0.946)	
BDAC1. Our company can easily integrate information from multiple sources	0.893
BDAC2. Our company can routinely use data visualization techniques to understand complex information	0.914
BDAC3. Our company can effectively decompose information to analysis the root cause of problems	0.938
BDAC4. Our company can easily deploy big data analytics applications to directors' devices	0.913
<i>Breakthrough innovation (BI)</i> ($\alpha = 0.845$, AVE = 0.683, CR = 0.846)	
BI1. Our product makes existing products obsolete	0.827
BI2. Our product incorporates a radically new technological knowledge	0.859
BI3. Our product significantly enhances customers' product experiences	0.800
BI4. The use of our product requires a major learning effort by mainstream customers	0.818

Source(s): Authors' work

Table 2.
Reliability and
convergent validity

we controlled R&D intensity, and measured it by using the ratio of R&D investments to sales. Fourth, we also considered digitalization experience as it positive affects a firm's digital capability, and measure it by using the years a firm has adopted digital technologies (Gupta and George, 2016).

3.3 Bias and common method variance

This study used *t*-test in firm size, firm age, R&D intensity and digitalization experience to examine non-response bias and later-response bias (Gerbing and Anderson, 1988). According to the measurements, these variables were operationalized as continuous variables. We respectively compared the means of these variables between respondents and non-respondents, early respondents and later respondents by SPSS. Results indicated no significant differences between respondents and non-respondents in firm size ($p = 0.482$), firm age ($p = 0.365$), R&D intensity ($p = 0.359$), digitalization experience ($p = 0.970$). Moreover, there were no significant differences in firm size ($p = 0.427$), firm age ($p = 0.664$), R&D intensity ($p = 0.243$), digitalization experience ($p = 0.925$) between early respondents and later respondents. Hence, neither non-response bias nor later-response bias is a serious concern in this study.

To assess the potential common method variance (CMV) deriving single source, this study compared two nested models through confirmatory factor analysis (CFA) by *Smart-PLS* (Podsakoff *et al.*, 2012). One model is the substantive factor model, in which items were loaded on their own research variables. The other is the common method factor model, in which a latent common method factor including all items were added in the substantive factor model. As shown in Appendix, the results show that the average explained variance of the items in the substantive factor model is 0.712, and 0.713 for the common method factor model. That is, the common method factor averagely explains 0.001 of total variances, indicating a very small magnitude of CMV. Furthermore, almost loadings on common method factor are insignificant, suggesting an insignificance of CMV. Additionally, we also performed Harman's single-factor test which entails composite-based and factor-base analyses (Kock, 2021). In Harman's single-factor test, a model with a single latent variable was created, and the total variance explained is in fact the average variance extracted (AVE). An AVE value greater than 0.5 indicates the presence of CMV. We performed composite-based PLS test and factor-based PLS test by *WarpPLS*. The AVE of each test is 0.400 and 0.361, both are less than 0.5. Therefore, CMV is not a major threat in this study.

3.4 Reliability and validity

The reliability of research variables was evaluated via construct reliability by *Smart-PLS*. Hair *et al.* (2019) suggested that the adequate construct reliability should meet the requirements that both Cronbach's α and composite reliability (CR) exceed 0.7. Table 2 reports that Cronbach's α and CR of each research variables were greater than the minimum threshold, resulting in an acceptable construct reliability.

The validity of research variables was tested in terms of convergent validity and discriminant validity by *Smart-PLS*. For convergent validity, this study calculated item loadings and average variance extracted (AVE) of each research variables. As shown in Table 2, all the item loadings and AVE of each research variables were greater than the minimum threshold 0.5 (Fornell and Larcker, 1981). Therefore, the convergent validity was supported.

For discriminant validity, this study used two methods suggested by Hair *et al.* (2019) by *Smart-PLS*. First, we performed heterotrait-monotrait (HTMT) ratio analysis to compute the HTMT values. An HTMT value below 0.9 indicates a satisfactory discriminant validity. As shown in Table 3, all the HTMT values were lower than 0.9. Second, we compared each

variable's AVE square root with correlation coefficients between variables. Results in Table 4 shows that, each variable's AVE square root was greater than the highest correlation coefficient (Fornell and Larcker, 1981). Taken together, results in Tables 3 and 4 provide a strong evidence of discriminant validity.

4. Analysis and results

4.1 Normality, descriptive statistics and correlations

To ensure the normality of research data, we performed a non-parametric test, namely Mardia test (Byrne, 2016). If $| \text{std-MK} |$ is less than 5, the data conform to a multivariate normal distribution, where std-MK is the standardized Multivariate Kurtosis. The results report the $| \text{std-MK} |$ is 2.45, indicating the normality of research data. Moreover, to describe the features of the research data and further perform hypotheses testing, this study conducted descriptive statistics and correlations analyses by SPSS. Table 4 reports the descriptive statistics and correlation coefficients of variables in this study. The results suggested that there are positive correlations among key research variables.

4.2 Hypotheses testing

To mitigate the collinearity, this study centered all the variables (Hair et al., 2019). As shown in Table 4, the VIF of each research variable was lower than the maximum threshold of 3. Hence, this study has no serious collinearity issue. To test the theoretical model and hypotheses, this study adapted regression analysis by SPSS. To isolate the effects of independent variables on the dependent variable, this study included demographic characteristics, such as firm size, firm age, R&D intensity, and digitalization experience. This also ensures that the observed relationships are not confounded by demographic differences among respondents.

4.2.1 Direct effects. H1a states that forward-looking search has a positive effect on breakthrough innovation. The linear direct effect can be expressed as Equation (1) (Igartua and Hayes, 2021). Model 2 in Table 5 reports that forward-looking search is significantly and positively related to breakthrough innovation ($\beta = 0.229, p < 0.001$). Hence, H1a is supported.

$$Y = \beta_0 + \beta_1 X$$
 Equation(1)

H1b states Backward-looking search has an inverted U-shaped effect on breakthrough innovation. This study used a three-step procedure proposed by Hanns et al. (2016) to examine the inverted U-shaped relationship. An inverted U-shaped relationship between independent variable X and dependent variable Y can be expressed as Equation (2). Its establishment should meet the following conditions. First, β_2 should to be significant and negative. Second, the slope needs to be steep at both ends of the X-range. In detail, the slope at the low end, supposed as X_L , which is $\beta_1 + 2\beta_2 X_L$, must be significant and positive. The slope

Variables	FLS	BLS	BDAC	BI	BDAC*FLS	BDAC*BLS
FLS	–					
BLS	0.656	–				
BDAC	0.327	0.243	–			
BI	0.636	0.676	0.428	–		
BDAC*FLS	0.450	0.429	0.039	0.170	–	
BDAC*BLS	0.431	0.520	0.036	0.342	0.445	–

Source(s): Authors work

Table 3.
Heterotrait-
monotrait ratio

Table 4.
Descriptive statistics
and correlations

Variables	Mean	SD	VIF	1	2	3	4	5	6	7	8
1. Firm size	2.838	1.159	1.083	–							
2. Firm age	3.079	1.132	1.034	0.098	–						
3. R&D intensity	3.025	1.026	1.060	0.179***	0.056	–					
4. Digitalization experience	3.073	1.129	1.050	0.157**	–0.072	–0.073	–				
5. FLS	3.485	0.797	1.498	0.041	–0.060	0.053	–0.036	0.808			
6. BLS	3.465	0.858	1.440	–0.013	–0.069	0.081	–0.021	0.545***	0.821		
7. BDAC	3.265	1.606	1.118	–0.019	–0.113*	0.080	0.039	0.290***	0.216***	0.915	
8. BI	3.385	0.886	–	–0.033	–0.156**	0.067	–0.059	0.531***	0.569***	0.381***	0.826
Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Numbers in italic on the diagonal indicate the square root of AVE											
Source(s): Authors' work											

	BI					
Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Control variables</i>						
Firm size	−0.021	0.002	0.004	0.013	0.006	0.015
Firm age	−0.162**	−0.096*	−0.081	−0.083	−0.074	−0.073
R&D intensity	0.076	0.035	0.020	0.023	0.021	0.024
Digitalization experience	−0.062	−0.033	−0.043	−0.036	−0.053	−0.046
<i>Independent variables</i>						
FLS		0.229***	0.193***	0.227***	0.209***	0.248***
BLS		0.240***	0.242***	0.272***	0.238***	0.262***
BLS square		−0.343***	−0.311***	−0.307***	−0.256***	−0.228***
<i>Moderator variables</i>						
BDAC			0.176***	0.157***	0.118*	0.086
<i>Interactions</i>						
FLS × BDAC				0.107*		0.135**
BLS × BDAC					0.053	0.025
BLS square × BDAC					0.134*	0.169*
R^2	0.035	0.483	0.510	0.519	0.517	0.529
R^2 change	—	0.448	0.027	0.009	0.007	0.019
F	2.784*	40.843***	39.693***	36.400***	32.386***	30.811***

Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Source(s): Authors' work

Table 5.
Regression results

at the high end, supposed X_H , which is $\beta_1 + 2\beta_2 X_H$, must be significant and negative. Third, the turning point needs to be within the X-range.

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 \quad \text{Equation(2)}$$

In model 2, backward-looking search squared is significantly and negatively related to breakthrough innovation ($\beta = -0.343$, $p < 0.001$). The range of standardized backward-looking search is $[-2.465, 1.535]$. The slope at -2.465 , which is 1.931, is significant and positive, whereas the slope at 1.535, which is -0.813 , is significant and negative. Moreover, the turning point of the relationship between backward-looking search and breakthrough innovation is 0.350, which is within the X-range. Taken together, [H1b](#) is supported.

4.2.2 Moderating effects. [H2a](#) states that big data analytics capability moderates and strengthens the positive effect of forward-looking search on breakthrough innovation. The moderation can be expressed as [Equation \(3\)](#) ([Igartua and Hayes, 2021](#)). Model 4 in [Table 5](#) reports that the interaction of forward-looking search and big data analytics capability is significant and positive ($\beta = 0.107$, $p < 0.05$). Furthermore, in Model 6, the coefficient of the interaction is still significant and positive ($\beta = 0.135$, $p < 0.01$). Hence, [H2a](#) is supported.

$$Y = \beta_0 + \beta_1 X + \beta_2 XM \quad \text{Equation(3)}$$

[H2b](#) states that big data analytics capability moderates and flattens the inverted U-shaped effect of backward-looking search on breakthrough innovation. The moderation in an inverted U-shaped relationship can be expressed as [Equation \(4\)](#), in which M is the moderator. The establishment of moderation depends on the shift of turning point and the change of shape in the inverted U-shaped curve ([Hanns et al., 2016](#)). For the turning point, a positive $\beta_1\beta_4 - \beta_2\beta_3$ indicates a rightward movement, while a negative $\beta_1\beta_4 - \beta_2\beta_3$ indicates a leftward movement. For the shape, a significant and positive β_4 indicates a flattening change, while a significant and negative β_4 indicates a steepening change.

Based on the results in Model 5, this study computed the shift of the turning point in the inverted U-shaped relationship between backward-looking search and breakthrough innovation, which is 0.350. The positive shift indicates a rightward movement as big data analytics capability increases. As the coefficient for the interaction between backward-looking search square and big data analytics capability is significant and positive ($\beta = 0.134$, $p < 0.05$), big data analytics capability flattens the inverted U-shaped relationship between backward-looking search and breakthrough innovation. Additionally, results in Model 6 also confirm a rightward movement and a flattening change of the inverted U-shaped relationship between backward-looking search and breakthrough innovation. Hence, H2b is supported.

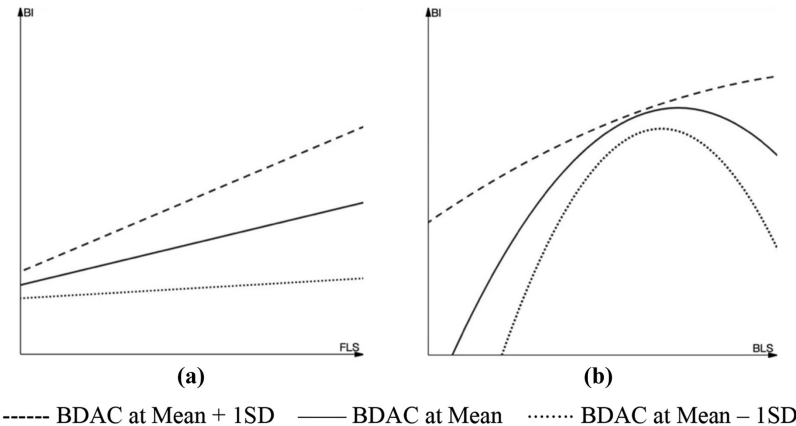
To depicts the moderating effects of big data analytics capability, we respectively plotted the forward-looking search—breakthrough innovation relationship and the backward-looking search—breakthrough innovation relationship at the mean of big data analytics capability as well as minus and plus a standard deviation from the mean, which describe low, moderate and high level of big data analytics capability. In Figure 2(a), the linear curve between forward-looking search and breakthrough innovation is showing steep trends as big data analytics capability increases. This indicates that big data analytics capability can strengthen the positive effect of forward-looking search on breakthrough innovation. In Figure 2(b), at high degree of big data analytics capability, the inverted U-shaped curve between backward-looking search and breakthrough innovation is flattened and becomes almost linear. This concludes that big data analytics capability can effectively mitigate the negative effect of backward-looking search on breakthrough innovation.

5. Discussion and implications

5.1 Discussion

The purpose of this study is to investigate how boundary-spanning search affects breakthrough innovation from the perspective of search orientation and examine the moderating role of big data analytics capability. Based upon empirical analyses on data from China’s advanced manufacturing firms, this study draws the following findings.

Figure 2. Moderating effects of big data analytics capability on forward-looking search – breakthrough innovation relationship (a) and backward-looking search – breakthrough innovation relationship (b)



Source(s): Authors work

First, forward-looking search has a positive effect on breakthrough innovation. Firms that conduct forward-looking search frequently scan for external knowledge to capture market trends and potential opportunities for new products. Such knowledge is the basis for firms to develop proactive strategy plans and think outside the box, thus generating novel insights for breakthrough innovation. It also enables firms to restructure interpretation procedure and create agile R&D routines to enhance breakthrough innovation. Hence, firms are able to think differently than before and capture novel signals of breakthrough innovation. This finding elucidates the influences of forward-looking search on breakthrough innovation, and highlights forward-looking search as a key driver of breakthrough innovation. It is also consistent with the arguments that forward-looking search plays a critical role in firms' innovation activities (Jissink *et al.*, 2019; Marinković *et al.*, 2022).

Second, backward-looking search has an inverted U-shaped effect on breakthrough innovation. Firms that conduct backward-looking search constantly look for external knowledge to improve existing products and processes. When the level of backward-looking search increases from low to moderate, breakthrough innovation increases as backward-looking search increases firms' motivation and risk tolerance to achieve breakthrough innovation. However, when the level of backward-looking search increases from moderate to high, it will constraints firms' search scope and obtain outdated knowledge, thereby hampering breakthrough innovation. This finding reveals that the positive effect of backward-looking search on breakthrough innovation initially increases, but after a certain point, it will decrease and finally disappear. Existing studies have inconsistent arguments about the effects of backward-looking search on breakthrough innovation. Some suggested the positive effects (Hu *et al.*, 2022), while others claimed the negative effects (Posen *et al.*, 2018). By taking the levels of backward-looking search into consideration, this finding provides a new perspective to understand the relationship between backward-looking search and firms' innovation.

Third, big data analytics capability respectively moderates the effects of forward-looking search and backward-looking search on breakthrough innovation. By enabling informed strategy plans, creative idea identification and cross-functional collaboration, big data analytics capability strengthens the positive effect of backward-looking search on breakthrough innovation. On the other hand, big data analytics capability makes the inverted U-shaped curve between backward-looking search and breakthrough innovation more linear by strengthening the positive effect and mitigating the negative effect of backward-looking search. This finding demonstrates the moderating roles of big data analytics capability in utilizing two types of searches to promote breakthrough innovation. It also supports the arguments of Hyun *et al.* (2023) and Mikalef *et al.* (2019) that big data analytics capability is an important accelerator in firms' innovation activities.

Given China's huge manufacturing industry, fast innovation and widely digital technology adoption, these findings may effectively capture the relationships among boundary-spanning search, big data analytics capability, and breakthrough innovation in the digital era. Similarly, these findings may be applied in various contexts where firms are striving to enhance their innovation capabilities and adopt new digital technologies, such as technology and IT industry, healthcare industry, energy and utilities industry.

5.2 Theoretical implications

Our findings contribute to extant research by three ways. First, this study extends the research on the relationship between boundary-spanning search and breakthrough innovation by taking search orientation into account. The rapid advancement of digital transformation requires a shift in firms' focus from unmet past needs to both latent future needs and unmet past needs. Although extant research has investigated the influences of

boundary-spanning search on breakthrough innovation from various perspectives (Wang *et al.*, 2020; Yang *et al.*, 2021), little paid attention to firms' search orientations. By highlighting the impacts of digital transformation on firms' search orientations, this study divides boundary-spanning search into forward-looking search and backward-looking search, and develops a theoretical model to elucidate their effects on breakthrough innovation. This not only enriches the literature of boundary-spanning search, but also advances the theoretical development on the relationship between boundary-spanning search and breakthrough innovation, thus extending RBV.

Second, this study uncovers the different effects of boundary-spanning search with different orientations on breakthrough innovation. A firm's search orientation partially determines its search efforts and outcomes (Spanjol *et al.*, 2012). However, there is no adequate understanding of how boundary-spanning search with different orientations affects breakthrough innovation. Our results confirm the positive effect of forward-looking search on breakthrough innovation and the inverted U-shaped effect of backward-looking search on breakthrough innovation. In this way, this study identifies the different influences of forward-looking and backward-looking searches on breakthrough innovation, and articulates the relationship between boundary-spanning search and breakthrough innovation from the perspective of search orientation. Moreover, by revealing the inverted U-shaped effect of backward-looking search on breakthrough innovation, this study also offers reasonable explanations to the inconsistent arguments on this relationship (Hu *et al.*, 2022; Posen *et al.*, 2018).

Third, by examining the moderating role of big data analytics capability, this study clarifies the crucial condition under which firms can increase the efficacy of boundary-spanning search in achieving breakthrough innovation. Recent studies highlighted big data analytics capability as an accelerator of firms' innovation processes (Hyun *et al.*, 2023; Kumar *et al.*, 2021), but whether and how it accelerates the transformation from boundary-spanning search to breakthrough innovation is still unclear. This study suggests that big data analytics capability strengthens the positive effect of forward-looking search on breakthrough innovation. What's more, as big data analytics capability increases, the inverted U-shaped curve between backward-looking search and breakthrough innovation is flattened and becomes almost linear. Hence, this study sheds light on the underlying mechanism by which boundary-spanning search can enhance breakthrough innovation, and provides a contingent viewpoint to deeply understand the relationship between boundary-spanning search and breakthrough innovation. Furthermore, this study also extends the research on boundary-spanning search and breakthrough innovation by including big data analytics capability as a moderator (Jung and Lee, 2016; Wang *et al.*, 2020; Yang *et al.*, 2021).

5.3 Managerial implications

This study has several managerial implications. First, managers should pay attention to the influence of digital transformation on firms' knowledge and innovation management. Digital technologies have revolutionized firms' knowledge and innovation management by enhancing external knowledge access, enabling open innovation, facilitating communication and collaboration and so on. To improve the efficiency of boundary-spanning search, firms can leverage various digital technologies. For example, firms can subscribe to industry experts' blogs or social media accounts to obtain updated insights and suggestions from them. It can also be helpful to employ specialized technology tools and intelligence services to track competitor, marketing trends and emerging technologies. Additionally, firms can participate in open innovation platforms and ecosystems to connect with other firms, innovative start-ups and research institutes.

Second, managers should be aware of the differences resulting in different search orientations. On the one hand, firms should foster corporate foresights in boundary-spanning search to capture future needs and trends. Managers can regularly participate in industry seminars and research reports, and pay attention to industry trends and emerging technologies. Reading industry news, communicating with peers and practitioners may also help to broaden managers' thinking to anticipate future changes. On the other hand, when searching knowledge for unmet past needs, managers should control the search activities at an appropriate level. Managers can set the goal of their search activities according to firms' current demands, and develop corresponding indicators to measure the activity effectiveness. Moreover, managers can utilize digital tools and analytics to monitor and evaluate the effectiveness, and make timely adjustments about search strategies and search methods.

Third, managers should take several steps to improve firms' big data analytics capabilities. The key to improving firms' big data analytics capabilities is to build an efficient data pipeline to collect, clean, integrate, and reserve data from various resources. It is also important to adopt appropriate big data analytics tools and techniques according to firms' demands and data size. In addition, managers should place emphasis on developing data teams with specialized knowledge and skills. These teams may include data engineers, data scientists, and business analysts who can be responsible for data processing, modeling, and interpretation. Last but not least, given that big data analytics is an iterative process, managers should continuously monitor and evaluate analysis results. Based on feedbacks and data-driven insights, managers can make adjustments and optimizations to improve the effectiveness and value of data analytics.

Moreover, this study also has social contributions regarding to the United Nations Sustainable Development Goals (SDGs). One of the SDGs is Industry, Innovation, and Infrastructure: build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation. By investigating the mechanism underlying the relationship between boundary-spanning search and breakthrough innovation in the digital era, this study provides insights about how organizations can leverage boundary-spanning search to drive innovation and technological advancements. Since boundary-spanning search often involves collaboration and partnerships with innovation actors, this study may also facilitate the development of global partnerships that can enhance innovation and deal with complex challenges, thus promoting sustainable development.

6. Conclusion, limitations and future research

The rapid advancement of digital transformation requires a shift of firms' focus from unmet past needs to both latent future and unmet past needs. However, how boundary-spanning search with future orientation and past orientation affects breakthrough innovation remains unclear. From the perspective of search orientation, this study divides boundary-spanning search into forward-looking search and backward-looking search. Based upon RBV, this study develops a theoretical model to explore the effects of forward-looking search and backward-looking search on breakthrough innovation, and tests the moderating role of big data analytics capability. Data from China's advanced manufacturing firms were used to examine the theoretical model and research hypotheses. The empirical results confirm that with the increase of forward-looking search, firms' breakthrough innovation increases, and big data analytics capability can accelerate the transformation from forward-looking search to breakthrough innovation. Furthermore, with the increase of backward-looking search, the likelihood that firms achieve breakthrough innovation will initially increase, but then decrease. Interestingly, big data analytics capability can mitigate the negative effect of backward-looking search, and increase the efficacy of the transformation from backward-looking search to breakthrough innovation. Our findings provide some theoretical and managerial contributions for firms' innovation in the digital era.

There also have some limitations in our study. First, our empirical results were mainly based on the data from advanced manufacturing firms in China, so whether the findings are applicable in other industries or countries remains to be verified. Therefore, future research can examine those findings in other industries or countries to discuss the possible cross-cultural differences. Second, the data used were mainly demographic data, and we didn't develop joint analyses between this data. Future research can consider more about the influence of demographic data on regressive analyses, and perform joint analyses to increase the research validity. Third, by uncovering the moderating role of big data analytics capability, this study may not fully capture the influence of firms' capabilities on innovation. As digital technology advances by leaps and bounds, there emerge some new developments in firms' capabilities, such as digital platform capability (Liu *et al.*, 2023). It will be meaningful for future research to explore the moderating roles of other emerging capabilities.

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Appendix

Construct	Indicator	Substantive factor loading (R ₁)	R ₁ ²	Common method factor loading (R ₂)	R ₂ ²	ΔR ²
FLS	FLS 1	0.816***	0.665	−0.057	0.666	0.001
	FLS 2	0.840***	0.705	0.071	0.707	0.002
	FLS 3	0.788***	0.620	−0.098	0.624	0.004
	FLS 4	0.788***	0.620	0.080	0.623	0.003
BLS	BLS 1	0.808***	0.653	−0.011	0.654	0.001
	BLS 2	0.858***	0.737	−0.030	0.737	0.000
	BLS 3	0.787***	0.619	−0.007	0.619	0.000
	BLS 4	0.830***	0.688	0.047	0.689	0.001
BDAC	BDAC 1	0.896***	0.802	−0.037	0.803	0.001
	BDAC 2	0.917***	0.841	−0.005	0.841	0.000
	BDAC 3	0.931***	0.867	0.082*	0.871	0.004
	BDAC 4	0.916***	0.839	−0.043	0.840	0.001
BI	BI 1	0.827***	0.685	0.002	0.685	0.000
	BI 2	0.859***	0.739	−0.005	0.739	0.003
	BI 3	0.801***	0.641	0.001	0.641	0.000
	BI 4	0.818***	0.669	0.002	0.669	0.000
Average		0.843	0.712	0.041	0.713	0.001

Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Source(s): Authors' work

Table A1.
Common method
variance analysis

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